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- CONCEPTS OF RISK
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Concepts of Risk: An Interdisciplinary Review

Part 1: Disciplinary Risk Concepts

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Risk is often defined as a function of magnitude and probability. However, risk concepts must also look at the context: Can the risk be quantified at all? Can people choose whether to take the risk? What are the benefits of taking it? Different approaches focus on different questions.

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Part 1: Disciplinary Risk Concepts
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Abstract

Risks, from natural or human sources, shape our physical, natural, and human environments. Most people demand healthy and safe products, trust that technologies that are meant to serve them do not act against their vital interests, and expect regulatory agencies to reduce risks to levels deemed tolerable by the majority of people. However, the scientific foundation for understanding and controlling risks is still far from being established. The main reason for this open situation is the multidisciplinary nature of risk. Risk means something different in the natural, engineering, psychological, social, and cultural disciplines. The first part of this review is an attempt to summarize the main disciplinary approaches to understanding and analyzing risk. It also seeks to provide some overarching categories that may help risk professionals in the academic and practical (risk management) fields to familiarize themselves with the fragmented field of risk studies and approaches, and to use the insights from these approaches in a more coherent and consistent fashion.

Keywords

concepts of risk, probabilistic risk assessment (PRA), risk analysis, risk assessment, risk governance, risk perception, social theories of risk

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Risk has been a focal topic of many disciplines, professional activities, and practical actions. Areas in which risks are being addressed range from natural hazards, technological threats, working conditions, ambient health impacts, crime, terrorism, and pollution to leisure activities. This article reviews the variety of concepts of risk in different disciplines and application areas with a specific emphasis on the social sciences.

All concepts of risk have one precondition: the contingency of human actions. At any time, an individual, an organization or a society as a whole faces several options for taking action (including doing nothing), each of which is associated with potential positive or negative consequences. Thinking about risks helps people to select the one option that promises more benefit than harm compared to all other options. If the contingent nature of our actions is taken for granted, the term "risk" denotes the likelihood that an undesirable state of reality (adverse effects) may occur as a result of natural events or human activities (definition originally in Kates et al. 1985, p. 21). This definition implies that humans can, and will, make causal connections between actions (or events). Consequences can be altered either by modifying the initiating activity or event, or by mitigating the impacts. The definition of risk[1], therefore, contains three elements: outcomes[2] that have an impact upon what humans value; the likelihood of occurrence (uncertainty); and a specific context in which the risk may materialize.

1 It is important to distinguish between risks and hazards. Hazards describe the potential of a technology, an event or an activity to cause harm to what people value. A hazard may be the toxicity of a chemical substance. A hazard becomes a risk if there is a likelihood that this potential is released in a way that it produces harm. Taking the example of the toxic chemical substance again, unless a person may be exposed to this substance and suffer the toxic effects there is no risk. The properties of a risk agent or event are necessary conditions for a risk but without a chance of exposure a risk does not exist.

2 Outcomes can, in principle, be positive or negative, depending upon the values that people associate with them. I will focus here on the negative outcomes of actions or events.

Consequently, three questions serve, in this essay, as structural tools for distinguishing the different concepts of risk and for discussing their present contributions and future challenges:

1. ***Scope of negative effects:*** What are undesirable outcomes, and who determines what undesirable means?
2. ***Conceptualization of uncertainty:*** How can we specify, qualify, or quantify the likelihood of undesirable outcomes?
3. ***Rule of aggregation for practical purposes in a given context:*** How do we aggregate outcomes, likelihood, and other risk-related factors into a common concept that allows risk comparisons, the setting of priorities, the inclusion of social or cultural context, and effective risk communication?

This review has been inspired by similar but incomplete or inconsistent attempts in the literature (Short 1984, Bradbury 1989, Renn 1992, Jaeger et al. 2001, McDaniels and Small 2004, Zinn and Taylor-Gooby 2006). Here, as well, it is impossible to provide a complete picture. In addition, the purpose of providing orientation is better served if similar approaches are pooled into one single category and if more peripheral attempts to capture risks are subsumed under one heading.

Risk Concepts in the Perspective of the Natural and Technical Sciences

Actuarial Analysis

The world of the insurance companies has a simple but very effective answer to the three questions. Their reference point is the expected value (i. e., the relative frequency of an event averaged over time, multiplied by the event's sum of insurance). The undesirable events are confined to physical harm to humans or ecosystems, which can be objectively observed or measured by appropriate scientific methods. The expected value of car accidents, for example, can be extrapolated from the statistical data about fatal accidents in previous years. This perspective of risk relies on two conditions: First, enough statistical data must be available to make meaningful predictions; second, the causal agents that are responsible for the negative effects must remain stable over the predicted time period (Cohen 1996). The resulting risk assessment is reduced to a single dimension representing a probability estimate averaging over space, time, and context.

Causal Models of Risks

The answers become more complex when the adverse effects cannot be observed as an immediate result caused by an agent (Kolluru 1995, Bedford and Cooke 2001). If we think of environmental risks, e. g., due to continuous or accidental release of carbon monoxide, benzene, or radioactive particles, the link between exposure and effect is often difficult to draw. In such risk assessments, causal relationships have to be explored through modelling. Models are not "miniature" representations of reality, but represent plausible and – in an ideal world – empirically confirmed representations of complex and non-obvious cause-effect relationships. In the field of medical risk, such models attempt to assess the likelihood of an impact of a specific agent on human health. Based on toxicological (animal experiments) or epidemiological studies (comparing a population exposed to a risk agent with a population not exposed), researchers try to identify and quantify the relationship between a hazard (such as acrylamide, chlorinated dibenzodioxins or ionizing radiation) and physical harm (such as cancer) in humans or other living organisms (WHO 1977, NRC 1991, Graham and Rhomberg 1996). Such risk assessments can serve as early warning signals to inform society that a substance may cause harm to humans or the environment, even if the effects are not obvious. In addition, dose-effect investigations help risk managers to define standards in accordance with observed or modelled threshold values. If there is no threshold value, as in the case of most genotoxic substances, risk assessments provide information about the probability of harm depending upon the dose. This information is vital for regulatory purposes.

Probabilistic Risk Assessment (PRA)

Another complication is experienced when people face technological risks (the possibility of technical malfunctions or human errors in handling machines). As a tool to model such failures and their consequences, experts use probabilistic risk assessments in an attempt to predict the probability of safety-relevant failures of complex technological systems, even in the absence of sufficient data for the system as a whole (Hauptmanns et al. 1987, Bedford and Cooke 2001).

Probabilistic risk assessments for large technological systems include tools such as fault and event trees, scenario techniques, distribution models based on geographic information systems >

(GIS), transportation modelling, and empirically driven human-machine interface simulations (IAEA 1995, Stricoff 1995, Kröger 2005). These tools have been developed to generate knowledge about cause-effect relationships, to estimate the strength of these relationships, to characterize remaining uncertainties and ambiguities, and to describe, in quantitative or qualitative form, other risk- or hazard-related properties that are important for risk management (IEC 1993). In short, probabilistic risk assessments specify what is at stake, calculate the probabilities for (un)wanted consequences and often aggregate both components into a single dimension by taking the integral over the loss-probability function (Kolluru 1995, p. 23 f.). More recent attempts avoid the aggregation of the two components and leave it to the risk evaluation or management team to draw conclusions from the juxtaposition of loss and probabilities (Kröger 2005). In addition, second-order uncertainties are introduced via confidence intervals to make the confidence of probability judgments more explicit (Apostolakis and Pickett 1998). Moreover, the evolution of probabilistic risk assessments over the last two decades has broadened the scope of outcomes to include monetary and quantifiable social losses as well as the results of complementary analyses such as human reliability investigations.

Several problems are associated with this approach to calculating the risks of accidents (IAEA 1995, Aven 2003, pp. 106 ff., Cullen and Small 2004). It has proven difficult to model common-mode failures (i. e., the simultaneous breakdown of technical components due to a common cause). Second, human-machine interactions are tricky to predict. They often rely on idiosyncratic events that defy systematic modelling. Similar problems arise when social events interact with technological systems, such as terrorist attacks, sabotage, social unrest, lack of safety culture, or the breakdown of organizational order (e. g., intoxicated air pilots or drug-using nuclear power plant operators). More recent attempts to model human factors in risk assessments include methods such as Monte Carlo Simulation, narrative scenario building and expert Delphis in order to provide more reliable estimates of the potential interfaces between social context and technical performance (Forester et al. 2006). In spite of the progress made in reliability analysis, there is still considerable doubt about the predictive power of these models.

Notwithstanding these difficulties, probabilistic risk assessments have been specifically valuable in detecting deficiencies in complex technical systems and in improving safety performance. Furthermore, they are powerful instruments to characterize the uncertainties in a technical system and to improve cost-efficiency of safety devices and measures.

Commonalities

These three approaches to risk (actuarial, model-based, and technological) have much in common. They anticipate potential physical harm to human beings, cultural artefacts, or ecosystems, average these events over time and space, and use relative frequencies or estimated probabilities (observed or modelled) as a means to specify likelihoods. The normative implication is obvious: since physical harm is perceived as an undesirable effect, technical risk analyses can be used to reveal, avoid, or modify the causes that lead to these unwanted effects. They can also be used to mitigate consequences, e. g., by accident management and emergency planning, if causes are yet unknown or impossible to change. Their instrumental functions in society are, therefore, oriented towards risk-based regulation, risk sharing, and risk reduction, through mitigation of consequences, standard setting, and improvements in the reliability and safety of technological systems. In addition, the data from technical risk analyses are crucial input for conducting risk-benefit analyses.

A Critical Review of the Technical and Natural Science Concepts

The technical analyses of risk have attracted massive criticism from the social sciences (Hoos 1980, Freudenburg 1989, Shrader-Frechette 1991, Adams 1995). First, what people perceive as an undesirable effect depends upon their values and preferences (Dietz et al. 1996). Second, the interactions between human activities and consequences are more complex, global and unique than the average probabilities used in technical risk analyses are able to capture (Fischhoff et al. 1982, Zinn and Taylor-Gooby 2006, p. 25). Third, the institutional structure of managing and controlling risks is prone to organizational failures and deficits that may increase the actual risk (Perrow 1999, Short 1984, Short and Clarke 1992). This interaction between organizational malfunctions and risk is usually excluded from technical risk analyses. Fourth, risk analysis cannot be regarded as a value-free scientific activity (Fischhoff 1995). Values are reflected in how risks are characterized, measured, and interpreted. Fifth, the – still widespread – numerical combination of magnitude and probabilities assumes equal weight for both components. The implication is indifference between high-consequence/low-probability and low-consequence/high-probability events with identical expected values. However, people show distinct preferences for one or the other (Slovic 1987, Renn 1990). Most people prefer a risk that will kill a few people at a time, rather than a risk that kills many people at once. Furthermore, technical risk analyses can provide only aggregate data over large segments of the population and long time duration. Each individual, however, may face different degrees of risk depending upon the variance of the probability distribution (Nowotny and Eisikovic 1990, Cullen and Small 2004). A person exposed to a larger risk than the average may legitimate-

ly object to a risk policy based on aggregate calculations (Jasanoff 1993, p. 127). The extent to which a person is exposed to a specific risk also rests on lifestyle factors and individual circumstances, both of which are mostly unknown to scientists performing risk analyses.

How valid are the criticisms by social scientists? In my opinion, all the critical remarks are well taken and point to the problem that technical risk analyses represent a narrow framework that should not be the single criterion for risk identification, evaluation and management. Technical risk analyses rest on many conventions (Weinberg 1972, Renn 2005), such as the selection rules for identifying undesirable effects, the choice of a probability concept, and the equal weighting of probability and magnitude, which is still the most often practiced procedure for aggregation. All these conventions can be defended through logical reasoning; but they represent only parts of what individuals and society experience as risk.

This does not mean, however, that technical risk analyses are unnecessary or less relevant than broader concepts of risk. They do serve a major purpose. After all, people get hurt or are killed in accidents, in natural disasters, or due to pollution (Shrader-Frechette 1991, p. 30). Technical risk analyses help decision-makers to estimate the expected physical harm. They provide the best knowledge about actual damage that is logically or empirically linked with each possibility of action (NRC 1983). Cancer, for example, may be caused by exposure to a harmful agent, such as benzene, and the effects are a function of the dose. In addition, for events that can be observed and repeated, probabilities are adequate tools to model their likelihood of occurrence in the future.

In terms of the three guiding questions stated above, technical analyses rely on relative frequencies or point estimates as a means of expressing probabilities. This concept aggregates data over space, populations, and time. The undesired effects are mostly confined to financial losses or physical harm to humans and the ecosystems, thus excluding most of the social and cultural impacts. There are attempts to include social value violation with the aid of economic valuation methods that will be described later, or by considering the costs for mitigation measures; yet most risk assessments concentrate on those impacts that are easy to measure and quantify. The limited scope of this approach constitutes both its weakness and its strength. Abstracting from the cultural context of risk-taking makes this concept of risk robust and universal. Physical harm and monetary losses may be the only consequences that (almost) all social groups and cultures agree to be undesirable (HMSO 1988, Kasperson 2005).

Looking at the prospects and limitations of technical risk analysis, it is obvious that the scientific communities in this field need to continue investing their efforts in order to improve the methodology for risk assessments. The main objectives are to standardize procedures and techniques, to enhance the spectrum of risk events that can be modelled, and to ensure that risk managers are able to understand and wisely use the instruments that risk analysts have developed over the last decades. Technical risk analysis has matured to become a sophisticated and powerful tool in coping with the potential harm caused by human actions or natural disasters (Morgan 1995).

The Economic Perspective: Emphasis on Utility

The concept closest to the technical approach is the economic concept of risk. The major difference here is the transformation of physical harm or other undesired effects into what economists have coined "utilities". The base unit of utility describes the degree of satisfaction, or dissatisfaction, associated with a possible action or transaction. The objective yardstick for measuring utility in classic economics is the amount of money that somebody is willing to pay for a change that provides a higher (lower) degree of benefit (loss) than remaining at the status quo. There are major debates today on whether the model of utility maximization is empirically valid (Jaeger et al. 2001, pp. 41 ff., Taylor-Gooby and Zinn 2006, pp. 8 f.). It is not clear whether humans actually distinguish different degrees of utility – a precondition for reconstructing cardinal utility functions. More recent work suggests that human choice behaviour depends more upon simplified heuristics that correspond with satisficing rather than optimizing strategies (Gigerenzer and Selten 2001).

The shift from expected harm to expected utility serves two major purposes. First, subjective (dis)satisfaction can be measured for all types of consequences, including symbolic, psychological, or social effects. Second, and more important, using utilities instead of physical harm provides a common "exchange rate" that enables each individual to compare options with different benefit profiles, according to overall satisfaction (Derby and Keen- >

ey 1981). The question, "How safe is safe enough?" cannot be answered by the technical concepts. The economic concept offers two main selection rules: a risk is acceptable if the corresponding benefit provides more utility than the risk distracts from the utility. Or, in case of concrete options: select the one option having the lowest risk among all options leading to identical benefits.

Several economists deduct the collective utility by looking at past behaviour (revealed preferences); others use surveys or auctions to determine the balance between the utility lost by implementing the risk, and the utility gained by receiving fringe benefits. Others again ask people directly, what they are willing to pay for imposing a risk on others, or what they are willing to accept for taking over a risk from others (expressed preferences).

Since risks denote possible costs rather than actual costs, consequences have to be weighted by the probability of their occurrence. Furthermore, since risks and benefits may not materialize until years after implementing the desired option, the consequences have to be discounted over time (Hyman and Stiftel 1988). Choosing the correct discount rate has been a major challenge to economists. Since market interest rates hardly reflect time preferences for collective risks, discount rates must be deliberately set according to theoretical reasoning or empirical surveys. There is also the ethical question of whether a loss of life or ecological damages can be discounted at all. Notwithstanding all these problems, economic theory attempts to integrate risk analysis as part of a larger cost-benefit consideration, in which risks are the expected utility losses resulting from an event or an activity. The ultimate goal is to allocate resources in order to maximize their utility for society (Shrader-Frechette 1991).

The economic risk concept constitutes a consistent and coherent logical framework for situations in which decisions are made by individuals, and in which decision consequences are confined to the decision-maker. In the risk area, however, both conditions are rarely met. First, most decisions on risks are collective decisions (public or meritocratic goods) which require the aggregation of individual utilities. How to aggregate individual utilities within a single societal welfare function remains an open problem in economics until this day. Second, many transactions between individuals imply the imposition of risks on third parties, who may not, or only marginally, benefit from the transaction itself (MacLean 1986). These problems are aggravated by the fact that utilities are often measured in monetary units, which is perceived as incommensurable with the risk of serious health impediments or death (Baram 1980). Additionally, some authors contest that risks can be expressed in utilities at all (compare the debate between Kelman 1981 and Butters et al. 1981). In spite of these criticisms, the economic approach serves several vital functions in risk policies (Pinkau and Renn 1998, pp. 271 ff.):

- It provides techniques and instruments to measure and compare utility losses or gains from different decision options, thus enabling decision-makers to make more informed (but not necessarily better) choices.
- It enhances technical risk analyses by providing a broader definition of undesirable and desirable outcomes which includes non-physical aspects of risk.
- Under the assumption that market prices (or shadow prices) represent social utilities, it provides techniques to measure distinctly different types of benefits and risks with the same unit. Such balancing of risks and benefits does not necessarily require monetary units. Alternative expressions, such as "quality-adjusted life years", may also serve as yardsticks for measuring the net utility balance.
- It includes a model for rational decision-making provided that the decision-makers can reach agreement about the utilities associated with each option.

In terms of the three guiding questions, the economic concept of risk uses a social definition of undesirable effects based on individual utilities, and treats these effects as real gains or losses to individuals or society. In contrast to the technical approaches, probabilities are not only conceptualized as relative frequencies (objective probabilities), but also as strengths of belief (subjective probabilities) (Fischhoff et al. 1981). Furthermore, people show different preferences when combining subjective probabilities and utilities, depending upon their basic attitudes towards taking risks (Kahneman and Tversky 1979).

The Psychology of Risk Perception: Subjective Processing of Risk Information

The psychological perspective on risk expands the realm of subjective judgement about the nature and magnitude of risks in three ways. First, it focuses on personal preferences, and attempts

TABLE: Intuitive biases of risk perception (after Rohrmann and Renn 2000, p. 25).

bias	description
availability	Events that come immediately to people's minds are rated as more probable than events that are of less personal importance.
anchoring effect	Probabilities are adjusted to the information available or the perceived significance of the information.
representation	Singular events experienced in person or associated with the properties of an event are regarded as more typical than information based on frequency of occurrence.
avoidance of cognitive dissonance	Information that challenges perceived probabilities that are already part of a belief system will either be ignored or downplayed.

to explain why individuals do not base their risk judgements on expected values (Lopes 1983, Luce and Weber 1986). One of the interesting results of these investigations was the discovery of consistent patterns of probabilistic reasoning that are well suited for most everyday situations. People are averse to risk if they face potential losses, and are risk-prone if they expect even small gains (prospect theory developed by Kahneman and Tversky 1979).

Second, more specific studies on the perception of probabilities in decision-making identified strong biases in people's drawing inferences from probabilistic information (Festinger 1957, Tversky and Kahneman 1974, Renn et al. 2007; see table).

Risk managers should be aware of these biases because they may be underlying causes for a public response. For example, the frequent media coverage about bovine spongiform encephalopathy (BSE), and a potential link to a certainly fatal human disease has alarmed the public and promoted a response of outrage based on the availability bias. Yet, the question remains: why do most people seem to underestimate the probability of contracting such a disease while amplifying the dread associated with the individual suffering from the disease? In order to understand this response, one needs to understand the semantic images that govern people's risk perception (Streffer et al. 2003, pp. 265 ff.). The literature lists in particular five semantic images that serve as powerful guides for individuals to navigate through an abundance of often contradictory information and provide an efficient method of balancing the time for collecting and processing information with the personal need for orientation and attitude formation. These semantic images refer to:

- ***Risk as a pending danger (fatal threat):*** In many areas of technology, particularly industrial technology, major accidents involving safety system failures can have catastrophic effects on humans and the environment. In technical safety philosophy, the main aim is to reduce the likelihood of such failure occurring, in order to ensure that the probability of impact is as small as conceivably possible. But the stochastic nature of such an event makes it impossible to foresee when such an event will actually occur. As a result, an event could, theoretically, occur at any time, although its likelihood is extremely low. Investigating the perception of these rare but possible catastrophic events shows that probability plays hardly any role: That the event can occur at any time frightens people and most likely will lead to rejection. Additionally, the risk-benefit distribution is perceived as inequitable. Risk sources in this category include major facilities, such as nuclear power plants, liquid natural gas (LNG) storage facilities, chemical production sites, and other human-made sources of potential danger that could have catastrophic effects on humans and the environment in the event of a serious accident.
- ***Risk as a stroke of fate:*** Natural disasters are usually seen as unavoidable events with catastrophic effects. They are seen as predetermined by fate, while technical risks are perceived as the consequences of decisions and actions. As opposed to the circumstances of technical risk, the random nature of the event is not the fear-triggering factor (because randomness involves fate and not the unforeseeable consequences of inappropriate action). Rather, the relative rarity of the event provides psychological reinforcement for risk denial.
- ***Risk as a personal thrill:*** In this understanding, people take risks in order to test their own strength and to experience triumph over natural forces or other risk factors. To pitch oneself against nature or one's competitors, and to overcome a self-contrived risk situation, are the major incentives in entering into the activity. It may well be that the challenges offered by modern safety-conscious society are too few so that the – often instinct-based – desire of people for adventure and risk goes unsatisfied. Instead, artificial situations are created to provide a calculable and (through personal effort) surmountable risk that individuals expose themselves to voluntarily. Risk as a thrill is such a dominant motivator that many societies have developed symbolic risk situations in the form of sports, games, speculation, investment, and the rules of power acquisition as a channel for the "kick" in overcoming danger, and to replace any possible negative outcome with symbolic penalties.
- ***Risk as a gamble:*** Risk perception involved in playing the lottery or games of chance involves losses or gains independent of the player's ability. Typically, there must be an asymmetry between risks and gains in games of chance. Psychologists have long conducted in-depth studies on risk behaviour in games of chance. Gambling often involves hidden distribution ideologies (like dead cert betting systems, lucky numbers, or fair shares) rather than an orientation towards an expected value. Some 47 percent of Americans believe that lucky numbers increase certain players' chances of winning. If, how- >

ever, the random element receives due recognition, the perceived concept of stochastic payout distribution comes closest to the technical risk model. People are normally able to perform such probabilistic reasoning, but only in the context of gambling, lotteries, financial investment, and insurance. Horse betting is an excellent example for this.

- ***Risk as an indicator of insidious danger (slow agents):*** In recent times, public debate has acquired yet another understanding of risk. Increasing reports on environmental pollution and its long-term impacts have forced scientific risk assessments to adopt a role as early warning indicators. In this risk perception model, scientific studies facilitate detection of lurking danger and the discovery of causal relationships between activities or events and their latent effects. Affected groups depend upon information provided by third parties because, as a rule, these risks cannot be perceived with the human senses. Trust is hence a key variable in the perception of risks in this category. If the informants are believed to be trustworthy, people are willing to use a balancing approach between risks and benefits and to assign trade-offs between the two. If they feel that the informants are not trustworthy they demand zero risk. If the answer is maybe, often peripheral factors become decisive for a judgement. These include hints or cues related to conflicts of interest (who pays the communicator?); similarity of values (does the communicator share my lifestyle or cultural background?); or affiliation to a highly esteemed reference group (does the communicator belong to a social group whom I deeply respect – for example, Nobel Prize winners?).

Some risk sources evoke more than one semantic image. Combinations of the slow agents and pending danger images are particularly interesting. For example, nuclear power is associated with large-scale accidents and slow deterioration of human health induced by low-dose radiation from normal operations. The same applies to many of the climate risks. Such combinations can act as risk amplifiers in public perceptions, leading to strong personal as well as social reactions.

Within the psychological domain of investigating risk perception, two major methodological approaches have been pursued: one through psychometric scaling exercises by which individuals rate risks on attributes such as voluntary, controllable, dreadful, familiar to them, or known to science (Slovic 1992); and the other by mental models that reconstruct the mental associations of respondents between the risk and related subjects, such as actors involved, context, and attitudes towards the source of risk (Morgan et al. 2001, Breakwell 2007). Risk perceptions differ considerably among social and cultural groups. However, it appears to be a common characteristic in almost all countries in which perception studies have been performed, that most people form their beliefs by referring to the nature of the risk, the cause of the risk, the associated benefits, and the circumstances of risk-taking (Renn and Rohrmann 2000).

In terms of the three guiding questions, the psychological perspective on risk includes all undesirable or desirable effects that people associate with a specific cause. Whether these cause-effect relationships reflect real dangers or gains is irrelevant. Scientific assessments influence the individual response to risk only to the degree that they are part of individual perceptions (Covello 1983). Relative frequencies or other (scientific) forms of defining probabilities are substituted by the strength of belief that people have about the likelihood of an undesirable effect occurring (Fischhoff et al. 1981). Both aspects are combined in a formula that normally puts more weight on the magnitude of the effects than on the probability of their occurrence. The main insight is, however, that effects and likelihood are enriched by context variables such as the degree of perceived personal control, the perception of a social rather than an individual risk, or the familiarity of the risk situation (Slovic 1992, Gould et al. 1988, Vlek 1996, Boholm 1998, Renn and Rohrmann 2000).

The focus on the individual and his/her subjective estimates is also the major weakness of the psychological perspective (Mazur 1987, Thompson et al. 1990, Wynne 1992, Jasanoff 2004). The number of the dimensions that people use to make judgements and the reliance on intuitive heuristics and anecdotal knowledge make it difficult, if not impossible, to aggregate individual preferences, and to find a common denominator for comparing individual risk perceptions. Risk perceptions vary among individuals and groups. Whose perceptions should be used to make decisions on risk? At the same time, however, these perceptions reflect the real concerns of people, and include those undesirable effects that the technical analyses of risk often miss. Facing this dilemma, how can risk perception studies contribute to improving risk policies? According to Fischhhoff (1985), they can:

- reveal public concerns and values;

- serve as indicators for public preferences;
- document desired lifestyles;
- help design risk communication strategies;
- represent personal experiences in ways that may not be available to the scientific assessment of risk.

In essence, psychological studies can help to create a more comprehensive set of decision options, and provide additional knowledge and normative criteria to evaluate them (Fischhoff 1994). Similar to the other perspectives, the psychological perspective on risk contributes valuable information for understanding risk responses and for designing risk policies, but it is limited in its comprehensiveness and applicability (Wynne 1984).

Social and Cultural Concepts of Risk: an Exercise in Diversity

A Taxonomy of Social Science Approaches to Risk

The risk world becomes even more complex with the sociological or cultural concepts of risks. The sociological perspectives include undesirable events that are socially defined and (in some cases) socially constructed. "Real" consequences are always mediated through social interpretation and linked with group values and interests (Bradbury 1989, Shrader-Frechette 1991, Wynne 1992, Luhmann 1993, O'Malley 2004). Possibilities for future events are not confined to the calculation of probabilities, but encompass group-specific knowledge and vision. Furthermore, possibilities are shaped by human interventions, social organizations, and technological developments (Freudenburg 1989, Short and Clarke 1992). Ignoring the connections between social organizations and technological performance may seriously underestimate or – under different circumstances – overestimate the likelihood of failures. Lastly, reality is seen as a system of both physical occurrences (independent of human observations) and constructed meanings with regard to these events as well as to abstract notions, such as fairness, vulnerability, and justice (MacLean 1986, Linnerooth-Bayer and Fitzgerald 1996).

What is missing, is a clear concept of how these factors influence social judgements about risks, individual behaviour, and institutional responses (see, for example, Hutter 2006). Notwithstanding the frustrations that are likely to evolve when sociologists and cultural anthropologists try to classify different schools of thought, the literature offers a wide variety of taxonomies, even in the narrow field of risk and disaster research (Kreps 1987, Bradbury 1989, Stallings 1990, Renn 1992, Short and Clarke 1992, Vlek 1996, Jaeger et al. 2001, McDaniels and Small 2004, Zinn and Taylor-Gooby 2006). They use different frameworks and classification criteria. All sociological and anthropological concepts of risk, however, do have in common the notion that "humans do not perceive the world with pristine eyes, but through perceptual lenses filtered by social and cultural meanings, transmitted via primary influences such as the family, friends, subordinates, and fellow workers" (Dietz et al. 1996, p. 46).

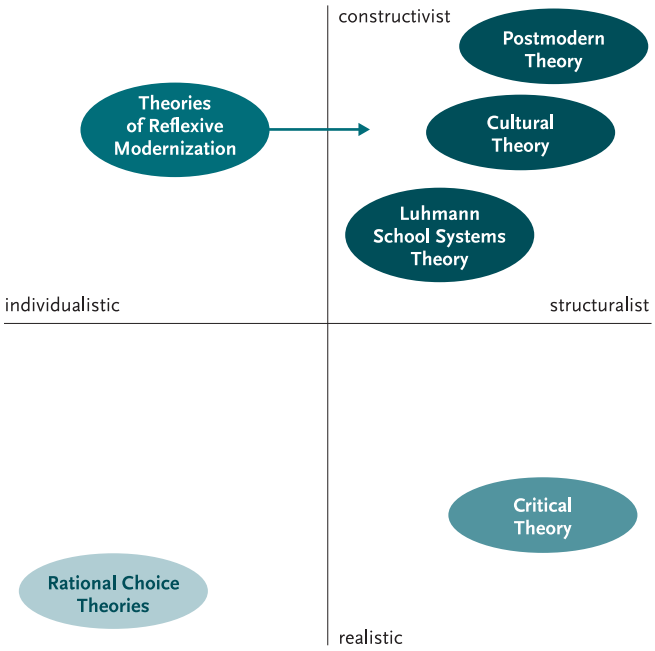

FIGURE 1: Classification of sociological approaches to risk. Source: modified from Renn 1992.

Rather than evaluate the existing reviews, this chapter ventures to add an additional taxonomy, shown in figure 1 (based on an earlier version in Renn 1992 and a similar approach in Zinn and Taylor-Gooby 2006). This taxonomy orders sociological and anthropological approaches with regard to two dimensions: individualistic versus structuralist (x-axis), and realistic versus constructivist (y-axis). The major reasons for this classification are as follows:

- The classification is simple and straightforward – and, thus, open to criticism.
- The classification allows using the questions of this comparative review as a structuring tool (constructivist versus realistic is associated with the question of whether uncertainties are framed as objective properties or as subjective expectations; individualistic versus structuralist is related to the context of aggregation).
- Most, if not all, social science concepts of risks can be grouped within the boundaries of the two dimensions.
- The two dimensions appear to be sufficient to distinguish between concepts that are clearly distinct from each other.

There are six social science based theoretical approaches to risk that are covered in this review (see a similar suggestion in Zinn and Taylor-Gooby 2006): the rational choice approach (Jaeger et al. 2001); the reflexive modernization approach by Beck (1986) and Giddens (2000); the systems theory approach of Luhmann (1993); the critical theory approach based on the seminal work of Juergen Habermas (1984, 1987); the post-modern perspective introduced by Foucault (1982) and further developed by Dean (1999) and others; and a cultural theory approach, originally in- >

troduced by Douglas (1966) and Douglas and Wildavsky (1982), recently represented by Adams (1995) and Lupton and Tulloch (2002). These approaches are described in more detail in the following sections.

The Rational Choice Approach

The notion of rational choice forming the philosophical basis for explaining risk has become the most popular and important theory of risk in economics (see p. 53) and many social sciences (Jaeger et al. 2001, pp. 20 ff.). In its widest form (as a global view), it parts from the assumption that human beings are capable of acting in a strategic fashion by linking decisions with outcomes. The more elaborate version of the idea of rational choice, the rational actor paradigm (RAP) makes a whole set of further claims that are more specific (Coleman 1990). The most important of these are:

- analytical separation of means and ends (people, as well as institutions, can, in principle, distinguish between ends and means to achieve these ends);
- maximization or optimization of individual utility (human actors select the course of action which promises to lead to more personal satisfaction than any other available one);
- predictability of human actions if preferences and subjective knowledge are known: rational actor theory is not only a normative model of how people should decide, but also a descriptive model of how people consciously or subconsciously select options and justify their actions.

The fundamental assertions linked to individual behaviour are also transferred to situations of collective decision-making or collective impacts of individual decisions. Organizations, for example, are treated as "virtual" individuals who act in accordance with the institutionalized preferences of the organization and its inherent rationale of selecting the most efficient means for reaching predefined goals (Jaeger et al. 2001, pp. 144 ff.). Underlying the individual as well as the collective interpretation of RAP is the basic assumption that all human actions can be described as problems of maximization or, in more contemporary versions of RAP, optimization.

The rational actor concept is located at the left side of our diagram (figure 1): the approach starts with the individual and rests on the assumption that risks describe the potential for real losses experienced by individuals over time.

In terms of the three leading questions, RAP addresses outcomes as subjective expectations that individuals link with different consequences of decision options. The uncertainties of these outcomes are captured by the strength of subjective belief that these outcomes may or may not materialize. The aggregation of likelihood and outcome is also modified by personal preferences or values of organizational culture. There are no universal yardsticks to evaluate risks, the evaluative judgment is always specific for the person or organization that makes the decision. A general agreement on risk, for example, can only be accomplished by searching for win-win situations where all individuals can gain additional benefits or by providing compensation to those who might suffer dis-benefits (Kaldor-Hicks Criterion). The rational actor concept comes closest to the economic understanding of risk but is also compatible with most psychological models of risk-taking.

The Theory of Reflexive Modernization and the "Risk Society"

The reflexive modernization approach combines the macro and micro levels and goes back to the seminal works of Ulrich Beck (1986) and Anthony Giddens (1990, 2000). It rests on the assumption that the meta-rationality of modernity (i. e., instrumental rationality, efficiency, justice through economic growth, and steady improvement of individual living conditions through scientific and technical progress) has lost its legitimizing power. This new development of disenchantment is caused by several developments within the transition from classic to reflexive modernity (cf. Lash 2000, Knight and Warland 2005, Zinn and Taylor-Gooby 2006, pp. 39 ff.):

- ***Individualization of lifestyles and social careers.*** Individuals have more options than ever before; but they lack social orientation and ontological security when faced with behavioural or moral dilemmas.
- ***Pluralization of knowledge camps and values.*** Contemporary societies are characterized by parallel systems of competing knowledge claims, moral judgement codes and behavioural orientations.
- ***Lack of overarching objectives and goals.*** All collective actions are challenged as being interest-driven or incongruent with somebody's beliefs, values, or convictions. This leads to the need for more legitimization; but there is hardly any reservoir for gaining legitimacy.
- ***Experience of negative side effects***. According to Beck the promises of modernization with respect to more progress and welfare have been undermined by the experience of negative side effects such as environmental degradation and inequitable distribution of resources.

In Beck's vision of reflexive modernization risk is the central focus of debate and controversy. He envisions five major characteristics of risk in modern societies compared to risk-taking in traditional societies (Beck 1986):

- ***Transboundary effects.*** Modern risks transgress sectoral, social, national, and cultural boundaries. They may originate in one country or one sector, but will then proliferate into other areas and sectors (e. g., BSE).
- ***Globalizing impacts.*** Risks tend to affect everybody (e. g., climate change).
- ***Increase of penetrating power.*** Risks tend to penetrate and transform social and cultural systems significantly and to change social behaviour (e. g., genetically modified organisms in agriculture).
- ***Incalculable nature.*** Due to the lack of boundaries and the complex global consequences of taking risks, the instruments and tools for calculating risks are inadequate and inaccurate

so that even insurance companies are unable to calculate premiums that are proportional to the respective risks (e. g., nuclear power plants).

- ***Lack of accountability.*** Potential victims of risks are being unduly burdened without their consent and without any institution or person being accountable for future damage (e. g., car exhaust).

What does the theory of reflexive modernization add to our understanding of risk? Most notably, individuals respond to the trends of individualization, pluralization, and globalization by searching for genuine orientation (Jaeger et al. 2001, Knight and Warland 2005). Confused by the plurality of lifestyles and value systems, they look for mental anchors that can provide them with a feeling of security and stability (Mythen 2005). These primary needs are constantly threatened by risk producers who impose risks on all members of society (without their consent) and, hence, threaten to destroy the basis of their livelihood. Whereas Beck claims that risk assessors in science and government fuel this irritation, since they systematically underestimate real threats and rely on a methodology aimed at legitimizing ubiquitous exposure of society to incalculable risks, Giddens believes that a reflexive approach to performing risk analyses (i. e., interdisciplinary, inclusive, and integrative) constitutes the only way to overcome the current stage of confusion. Beck does not offer a solution to the problem other than the suggestion of placing more trust in sub-political civil society actors and refusing to license any technical facility that cannot insure its maximum losses (Beck 1999).

Risk in the understanding of reflexive modernization constitutes real and measurable losses (left side of figure 1) that major actors may not recognize as they are captive in their own social reference systems (therefore the positioning towards the structuralist side).

In terms of the three leading questions, the theory of reflexive modernization addresses outcomes as social constructions of those who are deliberating different options for actions. These constructions are based on social and cultural orientations rather than factual expectations. The uncertainties of these outcomes are captured by the strength of subjective belief and the doubt about the ability of any knowledge system to provide accurate risk assessments. The aggregation of likelihood and outcome is part of a larger process in which genuine uncertainty prevails about the questions of whether the desired outcome materializes in the first place and whether it is over-compensated by negative side effects. The idea of rational decision-making is an illusion in this view since modernity offers no firm anchor for making reliable and inter-subjectively valid predictions.

Systems Theory

Systems theory understands risk as a fundamental social construct that is closely linked to the particular rationalities of societal subsystems. According to this theory (in particular, referring to Niklas Luhmann), human societies are organized as a variety of self-referential or autopoietic (self-reproducing) systems which define their own reality as well as an image of the world outside (Luhmann 1984). Systems include functional entities such as the law, the economy, and the political hierarchy. All systems and sub-systems have generated special communication media (such as legal codes, money, and power) to reduce complexity. These media ensure internal order and provide the necessary exchange with external systems. The sustainability of social systems depends upon the ability to exchange media. Media form the (binary) code of interaction within and between systems.

An important new feature to emerge in post-modern systems is the system-specific construction of risks and dangers (Luhmann 1990). Luhmann does not deny the possibility that systems can produce "objective" knowledge about the world, but points out that each observer is imprisoned in a social system that provides constructed meaning, rationality, and identity (Luhmann 1984). This construction leads Luhmann to the distinction between risk and danger. Social systems and sub-systems define the conditions for externalizing or internalizing "threats to health and the environment". In the terminology of Luhmann, external threats to a system are called dangers, while internal (and thus manageable) threats are called risks. This seemingly trivial semantic distinction embeds a deep theoretical demarcation. It addresses the recurrent contradiction where people perceive human life as becoming increasingly riskier, while at the same time other indicators of life chances, such as life expectancy, continue to improve in all affluent and even most developing countries. From this perspective, threats have not necessarily increased in terms of lives lost. Rather, social systems have increasingly internalized external threats – i. e., dangers – and have, thereby, transformed >

them into risks. Fates, acts of God, random events, catastrophes of all sorts, or capriciousness of nature remain as constructs for describing dangers. Similarly to Beck and Giddens, Luhmann argues that the more social systems act to shape the future, the more dangers are internalized and, axiomatically, the more risks are "created". For example, changing climate – once thought nature's caprice – is now viewed as significantly shaped by humans and is, therefore, a risk.

Since those who create risks may expose others to dangers, there is always the potential for incongruence between the risk-takers and risk (or rather danger) bearers in Luhmann's perspective. To deal with this systemic dilemma, decision-makers (and, as a means for exculpation, public risk managers) use formal tools of risk analysis to define and legitimate the acceptability of risks, while those who are affected by the decision's concequences use perceptions of dread, cultural interpretations, or competing professional judgements as a justification for rejecting dangers. Each line of reasoning is incompatible with the other, and there is no communication medium available for reconciling these opposing positions on risk-taking. Luhmann (1993, p. 159) put it this way: "With respect to the risk perspective in the future, neither consensual agreement on facts nor on values will be of any help; on the contrary, they will further aggravate the conflict." As a consequence, the political system is unable to resolve risk conflicts. System functioning is, therefore, continually threatened, as is communication among systems.

Risks can only be understood as a construction within a systems logic and must defy any objective definition. Furthermore, these constructions are reflections of the systems, not of individuals (thus the location on the right-hand side of the diagram, see figure 1).

In terms of the three leading questions, systems theory stresses the importance of social or cultural frames for selecting outcomes and providing meaning to them. Depending on the reference system, risks refer to loss of money (economic system), power (political system), solidarity (social system), health (health care system) or legal security (legal system) among others. The impacts on these system-based risk targets cannot be translated into a common currency. Neither money nor "utils" as suggested by the economic risk concepts can serve as a common denominator for different types of risk experiences. The uncertainty of these outcomes is not generally given by scientific calculation or subjective beliefs but by the system-specific rules for resolving complexity and contingency. Correspondingly, each system provides its own rules for the aggregation of likelihood and outcome. Individual preferences are employed by psychic systems, scientific conventions by the scientific system. The equal weighting of both components is a typical rule in technical systems but not in the economic or legal system.

Critical Theory

Critical theory, particularly Jürgen Habermas's (1984, 1987) theory of communicative action and of communicative competence respectively (1984, 1987), is based on an extended criticism of modernity that emphasizes the contradictions and untoward consequences of advanced or late capitalism. Critical theory relies partially on a systems perspective, but assumes an overarching rationality that bridges the different rationalities of the social systems and the institutions in a pluralist society. Critical theory suggests that due to the decline in the Enlightenment belief in a universal rationality, new social norms and values need to be generated. The fundamental goal of these emergent elements of rationality is to provide collective orientations that do not conflict with personal aspirations and agency (Habermas 1991, McCarthy 1973). Similar suggestions have been proposed by the new communitarians (Etzioni 1991).

Risks have emerged as dominating phenomena demanding political intervention and management. Decisions by the political system, based as they are on the exercise of power – rather than, e. g., a fairness doctrine – perpetuate an inequitable distribution of risks. According to critical theory, the only viable solution to overcome this imbalance is to create a forum for open discourse, where all actors have the opportunity to argue their interests and where thus conflicts are resolved in an equitable and rational manner. The process of discourse must be fair, transparent, and truthful (Renn 2004).

In critical theory risks are seen as real phenomena caused by structural forces such as capitalist systems. These real risks are not always obvious to the observer but can only be reconstructed in collective discourses (therefore the left top position in figure 1).

Critical theory is not specific about the three guiding questions about outcomes, uncertainty, and aggregation. Its main message is that outcomes and side effects of human action can be addressed by a comprehensive discourse linking instrumen-

tal, normative, expressive, and communicative rationality. The world of RAP is only one element of this overarching rationality that includes socially rewarding experiences beyond individual utility. Altruistic considerations play a role as well as personal relationships and cultural bonds. In contrast to Beck's theory of reflexive modernization and Luhmann's systems theory critical theory assumes the possibility and fruitfulness of meaningful interaction between systems and between risk creators and risk bearers. Societies can effectively deal with complexity and contingency of risks if the settings and political conditions for discourse are appropriate. There is little mentioning about uncertainty in critical theory. One can assume, however, that uncertainties can be properly handled by methodological rigidity when it comes to factual uncertainty, by truthfulness and openness when it comes to behavioural uncertainty (what will people do?), and by consistence and ethical justification when it comes to normative uncertainty.

Post-Modernism

Post-modern theorists have developed an extreme form of constructivism: Knowledge, moral norms, and aesthetic values represent nothing more than power-induced constructions by groups who abuse claims of universality for influencing or manipulating others (Lyotard 1984, Smart 1996, Dean 1999, see overview in Jaeger et al. 2001, pp. 221 ff.). Independent of the question of whether an objective reality exists, post-modernists believe that all claims towards an objective world are guided by personal interests and group-specific reasoning. They want to reveal the hidden power motives behind claims of individuals and groups to enforce behavioural, moral, or cognitive norms on others.

Risk is not at the centre of post-modern thinking. Many post-modernists are radical individualists who believe that the individual is able to cope with contingencies and to arrive at the most appropriate balance of expected positive and negative outcomes. However, what is seen as risks and what as benefits, and to what degree, depends upon the framing of social forces. Tversky and Kahneman (1981, p. 453) "use the term 'decision frame' to refer to the decision-maker's conception of the acts, outcomes, and contingencies associated with a particular choice. The frame that a decision-maker adopts is controlled partly by the formulation of the problem and partly by the norms, habits, and personal characteristics of the decision-maker." The theory of framing has been adjusted to post-modern theory by Tagg (1992, pp. 170 ff.), who analyzes the discursive and institutional relations of power that frame meanings. Risks are thus part of the game to legitimize power. Disguised by allegedly scientific rationale, people are coerced into accepting nuclear power, genetically modified organisms, industrial agriculture, nanotechnologies, etc. Similar arguments have been raised from a feminist perspective: the way society frames and evaluates risks reveals hidden agendas about gender stereotypes and leads to inequalities between men and women (Harding 1992). Post-modern theory has eluded risk analysts to the importance of framing and cultural biases in assessing and evaluating risks. For example, the threat of terrorism is framed by political elites in a way that the likelihood is highly exaggerated (or the low probability of being affected downplayed) and the randomness of becoming a victim emphasized in order to legitimize restrictions of personal freedom whereas the same elites emphasize the low probability of an accident and the remoteness of becoming a victim when they try to legitimize nuclear power stations.

Obviously the post-modern understanding of risk is constructivist and structuralist (right-top position in figure 1).

In terms of the three leading questions, post-modernists take the most radical position. Outcomes can only be seen from the perspective of each individual. There is no universal rationality. Any knowledge claims from scientific or regulatory authorities are interest-driven activities to maintain power. Uncertainty judgements in their eyes lend themselves to creating fear or confidence thus manipulating individuals to subordinate themselves under an allegedly common cause. Evoking fear helps to demand solidarity against a common threat, referring to confidence leads people to increased complacency as they are told that more competent institutions make the "right" decisions on their behalf. Finally context determines the legitimizing power of aggregation rules that in essence are arbitrary but framed in ways that they justify collective actions or restrictions of individual freedom.

Cultural Theory of Risk

Similar to the post-modern thinkers, cultural theorists of risk treat risks as social constructs that are determined by structural forces in society. Issues such as health threats, inequities, fairness, control, and others cannot be determined by scientific analysis, but only reconstructed from the beliefs and rationalities of the various actors in society (Douglas and Wildavsky 1982, Rayner 1990, Thompson et al. 1990, Grendstad 2000). The fabric and texture of these constructions reflect both the interests and values of each group or institution in the various risk arenas and the shared meanings of terms, cultural artefacts, and natural phenomena among groups. Risk policies result from a constant struggle of all participating actors to place their view of risk on the public agenda and to impose it upon others.

Several sociologists and anthropologists have attempted to distinguish typical combinations of values, world views, and conviction to form what they call "cultural prototypes". These groups represent different cultural groups in society with specific positions on risk topics as well as corresponding attitudes and coping strategies. Four or five prototypes have been defined: entrepreneurs, egalitarians, bureaucrats, atomized individuals, and (in some publications) hermits (Thompson 1980). They differ in their degree of *group* cohesiveness (the extent to which someone identifies with a social group) and the degree of *grid* (the extent to which someone accepts and respects a formal system of hierarchy and procedural rules) (figure 2, p. 62).

Organizations or social groups belonging to the ***entrepreneurial*** prototype perceive risk-taking as an opportunity to succeed in a competitive market and to pursue their personal goals. They are characterized by a low degree of hierarchy and a low degree >

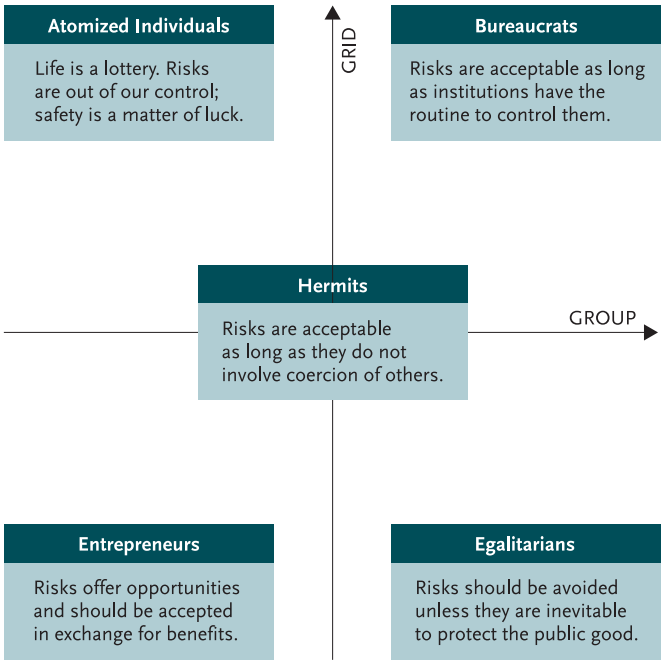

FIGURE 2: Cultural prototypes and their perspectives on risk. *Group* cohesiveness means the extent to which someone identifies with a social group, whereas the degree of *grid* stands for the extent to which someone accepts and respects a formal system of hierarchy and procedural rules. Source: adapted from Thompson et al. 1990 and Renn et al. 2007, p. 57.

of cohesion. They are less concerned about equity issues and would like the government to refrain from extensive regulation or risk management efforts.

This group contrasts with organizations or groups belonging to the ***egalitarian*** prototype, which emphasizes cooperation and equality rather than competition and freedom. Egalitarians are also characterized by low hierarchy, but have developed a strong sense of group cohesiveness, solidarity, and equity. When facing risks, they tend to focus on the long-term effects of human activities and are more likely to abandon an activity (even if they perceive it as beneficial) than to take chances.

The third prototype, the ***bureaucrats,*** relies on rules and procedures in order to cope with uncertainty. Bureaucrats are both hierarchical and cohesive in their group relations. They believe in the effectiveness of organizational skills and practices and regard a (risk) problem as solved when a procedure for its institutional management is in place.

The fourth prototype, the group of ***atomized*** or ***stratified individuals,*** as a matter of principle believes in hierarchy, but does not identify with the hierarchy it belongs to. These people trust only themselves, are often confused about risk issues and are likely to take high risks for themselves, but oppose any risk that they feel is imposed upon them. At the same time, they see life as a lottery and are often unable to link harm to a concrete cause.

In addition to the four prototypes, there may be a hybrid group called the ***autonomous individuals*** or the ***hermits,*** who can be grouped in the centre of the group-grid coordinates. Thompson (1980) describes autonomous individuals as self-centred hermits and short-term risk evaluators. They may also be referred to as potential mediators in risk conflicts, since they build multiple alliances to the four other groups and believe in hierarchy only if they can relate the authority to superior performance or knowledge.

Similar to post-modernism, cultural theory is based on the assumption that risks are socially constructed and caused by structural rather than individual forces (right-hand position in figure 1 with tendency towards structuralism).

In terms of the three leading questions, cultural theory of risk addresses outcomes as genuine expectations that different (sub)cultures associate with different consequences of decision options or events. In contrast to the post-modern theory or the systems theory the range of possible expectations in cultural theory is limited to a small set of cultural prototypes. These prototypes determine both: what is selected as a desirable or undesirable outcome and how uncertain or certain these outcomes are being perceived. The aggregation of likelihood and outcome is also modified by the dominant beliefs within each cultural prototype. In particular, context variables such as perceived equity play a major role in forming an overall judgement about the acceptability of risks.

Synopsis of Social Science Approaches

The social science perspectives on risk include a multitude of undesirable or desirable effects that people associate with a specific cause, leading to consequences for something that people value (Kasperson and Pijawka 2005, p. 31). Individuals or social/cultural groups respond according to their perception of risk and not according to an objective risk level or the scientific assessment of risk (Sjöberg 2006, p. 694). The main insight is that focusing on effects and likelihood alone is merely of limited use for social orientation. First, maximizing outcomes may not be the main objective of social actions in situations where overarching social goals – such as building solidarity, demonstrating empathy or strengthening one's identity – are pursued. Second, even if an individual perceives a situation as one of personal choice, the inability to predict outcomes, the interferences of one's own action with the actions of others, and the presence of competing and conflicting values and preferences make it difficult, if not impossible, to base one's own decisions on the evaluation of expected outcomes (Jaeger et al. 2001, pp. 247 ff.). Third, even if optimization of outcome is the preferred strategy, it is not only governed by magnitude and probability, but is also enriched by situational and risk-specific characteristics, such as the degree of perceived personal control, the perception of a social rather than an individual risk or the familiarity of the risk situation, or the experience of equitable risk-benefit distribution (Boholm 1998). Risk evaluation by individuals can, hence, be characterized by a functional relationship representing perceived violations of what humans value, perceived patterns of occurrence and social context variables. Fourth, most people rely on information from third parties when they are faced with unknown

risks. In this situation, they can either trust the information from experts or risk managers or not since they have no (empirical) way of proving one side wrong or the other. They have no lab at their disposal in order to investigate the risks of BSE, acrylamide, electromagnetic fields, or genetically modified plants. For risk professionals, it is, therefore, essential to build a trusting relationship with their target audiences. Trust, however, relies more on institutional arrangements, cultural traditions, and symbolic interactions than on factual evidence (Löfstedt 2005). As post-modern theory assumes trust is a powerful instrument for persuading people to accept decisions without having the capability to control the motivation and the level of evidence put forth by the organization that demands trust. Even if one does not accept the cynical world view of post-modern thinkers, it seems obvious that in complex and uncertain situations assigning trust to elites is on one hand unavoidable but on the other hand a risk in itself with respect to the actual competence and goodwill of these elites.

The various concepts within the cultural and sociological analysis provide some common, some conflicting, and some complementary answers to the three guiding questions. They all have in common that the definition of the desirability or undesirability of outcomes, the generation and estimation of possibilities, as well as the formulas to combine both aspects depend upon individual preferences, social context variables, and the cultural affiliation of the respective social group (Shubik 1991, Kasperson 2005). Society is experiencing an increase in pluralism of values, lifestyles, and knowledge systems (Beck 1986). Many philosophers and social scientists claim that communication between these pluralist groups has become more and more difficult and is likely to be impossible in the near future (Luhmann 1990). However, since the experience of risk spans the boundaries of confined social groups, communication is a prerequisite for risk management and policy implementation (Leiss 1996, OECD 2002). It is thus essential to provide semantic and organizational tools for creating a common language base among and between different groups, and to find new means of mediation and conflict resolution among the different stakeholders (Webler 1995). Critical theory might be one of the few approaches that is in principle optimistic about the prospects of such a communication. RAP also offers a perspective for common understanding if the parties agree to confine their discourse to facts (as revealed by scientific research) and interests (as being displayed in negotiations). All other social science concepts discussed in this article are pessimistic about the prospects of any overarching discourse between different camps and groups in society. The best one can hope for is a type of arrangement that people agree on as a minimal standard for enabling collective decision-making.

Résumé of the First Part

What are the major lessons to be learned from this review of disciplinary perspectives on risk? Are there opportunities to integrate technical, economic, psychological, and social science perspectives on risk, and how should one go about it? Our analysis shows that risk concepts of the technical and natural sciences address a limited scope of effects: objectively verifiable violations of what humans value, in particular harm to human health, environment, or wealth. Moreover, they confine likelihoods to numerical probabilities based on relative frequencies derived from experiments, models, expert judgements, and scenario techniques. This limited scope is a virtue as much as a shortcoming. Focused on real health effects, monetary losses, or ecological damage, these concepts are based on a societal consensus of (un-)desirability and a (positivistic) methodology that ensures equal treatment for all risks under consideration (Merkhofer 1984). The price society pays for this methodological rigor is the abstraction from the culture and context of risk-taking behaviour.

The economic, psychological, and social science perspectives on risk broaden the scope of expressing possibilities and likelihood, and expand the horizon of risk outcomes by referring to "socially constructed" or "socially mediated" realities. Naturally, they include (the perception of) actual damage, but are more focused on the evaluation of the risk context, the non-physical impacts, and the associations between the risk and social or cultural artefacts. If all society cared about was to reduce the amount of physical harm done to its members, technical analyses and some form of economic balancing would suffice for effective risk management. Included could be the perspective of organizational sociology to make sure that technical safety measures are paralleled by institutional control and monitoring (Perrow 1999, Freudenburg 1992). Psychology and the social sciences would only be needed to sell the risk management packages to the "mis- >

informed" public via risk communication. But society is not only concerned with risk minimization (Douglas and Wildavsky 1982, Leiss 1996). People are willing to suffer harm if they feel it is justified or if it serves other goals. At the same time, they may reject even the slightest chance of being hurt if they feel the risk is imposed upon them or violates their other attitudes and values (Fischhoff et al. 1985, MacLean 1986, Linnerooth-Bayer and Fitzgerald 1996). Context matters. So does procedure of decision-making, irrespective of outcome. Responsive risk management needs to take these socially mediated consequences into account.

What is needed, is therefore a comprehensive risk concept that spans the different perspectives and provides an integrated approach for capturing the physical and socio-cultural aspects of risk. Such a concept should be guided by the rigor and specificity of the technical and natural science approaches and inspired by the richness and plurality of the economic, psychological and social science approaches. Are there candidates that could be able to meet these two, on first sight contradictory, demands? This question will be addressed in the second part of this review to be published in the next issue of *GAIA*.

An extended version of this review will be published in the book *Risk Governance* (Renn in preparation). The sections on rational choice theory, reflexive modernization, and systems theory are partially based on a manuscript that I have written together with *Eugene Rosa* and *Aaron McCright* (Rosa et al. submitted).

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